Would a Bigger Heart Really Make the Grinch Strong?

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Could a small heart make someone grumpy? In Dr. Seuss's How the Grinch Stole Christmas, the Grinch starts with a heart two sizes too small. When he hears joyful singing from Whoville, his heart grows three sizes that day. But would a too-small heart really make someone grumpy, and what would happen if a heart went from too small to really big in an instant? The answer depends on what causes the heart to shrink or grow, experts say, and on whether or not you are a snake. Heart size is generally linked to body size, says heart doctor Michelle Kittleson, though some hearts are naturally bigger or smaller than average. There can be **downsides** to a too-small heart. In a 2025 study of more than 360,000 people in Australia, those with smaller-than-average hearts had a higher risk of dying within five years, perhaps because smaller hearts may not pump blood as well. Maybe the Grinch's heart was small because he did not exercise enough, Kittleson suggests, and that made him tired and grumpy. But would sudden growth be good for him? The human heart is a **four-chambered** pump, and there are many ways it can grow, notes **molecular biologist** Leslie Leinwand. With **vigorous** exercise, your heart gets bigger in a good way. Pregnancy also makes the heart grow, and if people stop exercising, or give birth, their hearts shrink back to normal size within months. But the Grinch's heart expanded in just moments, more like a snake's. Leinwand studies **pythons**, whose hearts can grow by up to 25 percent in a day after a large meal, and the bigger the meal, the more the heart grows.

Word Treasure

four-chambered -- A heart with four separate rooms that pump blood around the body.

molecular biologist -- A scientist who studies how living things work at the level of tiny molecules.

vigorous -- Strong and energetic, like exercise that makes you breathe hard.

pythons -- A kind of large snake.

downsides -- The bad or unwanted parts of something.

Background you need

The human heart is a four-chambered pump that moves blood around the body, carrying oxygen and nutrients to every part.

Heart size is usually linked to body size, so bigger bodies tend to have bigger hearts, though some hearts are naturally a bit bigger or smaller than average.

A heart can grow in a healthy way with vigorous exercise, and it can shrink back toward its normal size when exercise stops.

Some animals can change heart size much faster: a python's heart can grow quickly after a large meal, with a bigger meal leading to more growth.

Break it down

LEAD: The Grinch starts with a heart two sizes too small, then it grows three sizes when he hears joyful singing from Whoville, raising the question of whether a too-small heart really makes someone grumpy.

KEY EVENT: The Grinch's heart grows three sizes in a single day

+- WHO: The Grinch, heart doctor Michelle Kittleson, and molecular biologist Leslie Leinwand

+- WHAT: A heart that goes from too small to really big in an instant

+- WHEN: On the day the Grinch hears the joyful singing from Whoville

EVIDENCE: A 2025 study of more than 360,000 people in Australia linked smaller-than-average hearts to a higher risk of dying within five years, and pythons' hearts can grow by up to 25 percent in a day after a large meal.

OPEN QUESTION: Would sudden growth be good for the Grinch, since the answer depends on what causes the heart to shrink or grow, and on whether or not you are a snake?

Why it matters

Understanding how heart size relates to health helps explain why exercise matters for real hearts, not just storybook ones, and why sudden changes in the body can be risky.

Test what you remember

1. **How many sizes too small was the Grinch's heart at the start of the story?**
   - (A) Four sizes
   - (B) Two sizes
   - (C) One size
   - (D) Three sizes
2. **What happened when the Grinch heard joyful singing from Whoville?**
   - (A) He stopped hearing the singing
   - (B) His heart stopped beating
   - (C) His heart grew three sizes that day
   - (D) His heart shrank two sizes
3. **According to the 2025 study of more than 360,000 people in Australia, what was linked to smaller-than-average hearts?**
   - (A) A lower risk of dying within five years
   - (B) A heart that never changes size
   - (C) A heart that pumps blood better
   - (D) A higher risk of dying within five years
4. **How does heart doctor Michelle Kittleson describe the human heart?**
   - (A) A four-chambered pump
   - (B) A pump with no chambers at all
   - (C) A two-chambered pump
   - (D) A single-chambered pump
5. **What happens to a person's heart after they stop exercising or give birth?**
   - (A) It shrinks to two sizes too small
   - (B) It stays enlarged forever
   - (C) It grows even bigger
   - (D) It shrinks back to normal size within months
6. **By how much can a python's heart grow in a day after a large meal, according to Leslie Leinwand's research?**
   - (A) By up to 5 percent
   - (B) By up to 50 percent
   - (C) By up to 10 percent
   - (D) By up to 25 percent

Answer key (try the quiz first!): 1: B 2: C 3: D 4: A 5: D 6: D

Questions to think about

1. Michelle Kittleson: Heart size is generally linked to body size, and she suggests the Grinch's heart may have been small because he did not exercise enough, which made him tired and grumpy.

2. Leslie Leinwand: The human heart is a four-chambered pump that can grow in a good way with vigorous exercise or during pregnancy, and she studies pythons whose hearts grow quickly after a large meal.

MY THOUGHTS (at least 20 words):